Wall time per step

Grey: the step where the run died. Blank: step not available or not reached.

100k (69,412 cells)

	load	qc	normalise	hvg	pca	harmony	knn_snn	cluster	umap	tsne_default	tsne_matched	tsne_fft
bixverse mtx	11	0.9		0.5	1.5	4.7	1.6	1.3	2.6	26		10.0
bixverse h5ad	6.8	1.1		0.5	1.5	5.0	1.7	1.4	2.6	26		10
bixverse GPU*	6.8	0.7		0.3	0.5	2.1	1.0	1.0	0.6	26		9.6
Scanpy mtx	21	0.8	2.0	1.3	1.4	8.7	28	1.9	27	103		32
Scanpy h5ad	3.3	0.8	2.1	1.4	1.4	8.5	30	2.1	27	103		32
Scanpy dask	5.9	1.0	0.0	1.7	3.9	8.4	24	2.0	27	100		31
scrapper mtx	42	0.5	0.1	0.2	1.8	11	4.6	8.6	33	18	41	
scrapper h5ad	6.3	0.4	0.1	0.2	1.6	12	4.3	8.2	32	17	37	
Seurat std mtx	44	2.8	4.9	7.3	25	13	13	17	58	74		26
Seurat std h5ad	9.9	2.6	4.7	6.8	24	13	13	16	54	71		24
Seurat BPCells	15	1.2	2.2	2.7	75	14	14	18	57	78		26
OSCA HDF5	38	10	8.0	609	63	14	29	7.8	28	99		

500k (346,259 cells)

	load	qc	normalise	hvg	pca	harmony	knn_snn	cluster	umap	tsne_default	tsne_matched	tsne_fft
bixverse mtx	72	2.7		1.0	3.6	31	8.9	7.6	23	196		87
bixverse h5ad	29	3.6		1.0	3.5	33	9.5	7.5	24	191		86
bixverse GPU*	38	2.3		1.0	2.1	6.8	4.0	5.7	2.7	138		35
Scanpy mtx	OOM											
Scanpy h5ad	19	3.8	3.5	3.1	5.7	58	32	13	158	840		115
Scanpy dask	23	4.2	0.0	5.0	10	58	28	13	152	813		110
scrapper mtx	OOM											
scrapper h5ad	150	2.1	0.4	0.8	10	96	26	83	150	98	202	
Seurat std mtx	OOM											
Seurat std h5ad	OOM											
Seurat BPCells	75	4.8	8.3	9.2	357	92	90	111	344	629		109
OSCA HDF5	160	47	29	OOM (worker)								

full (684,932 cells)

	load	qc	normalise	hvg	pca	harmony	knn_snn	cluster	umap	tsne_default	tsne_matched	tsne_fft
bixverse mtx	112	4.5		1.8	6.9	69	19	17	51	507		272
bixverse h5ad	71	4.8		1.8	6.3	64	18	16	51	505		269
bixverse GPU*	71	3.0		1.8	6.2	17	8.3	12	6.0	324		90
Scanpy mtx	OOM											
Scanpy h5ad	OOM											
Scanpy dask	42	7.4	0.0	9.0	17	138	33	30	352	1,955		258
scrapper mtx	OOM											
scrapper h5ad	OOM											
Seurat std mtx	OOM											
Seurat std h5ad	OOM											
Seurat BPCells	140	9.2	16	16	703	220	201	249	749	1,600		227
OSCA HDF5	316	101	61	OOM (worker)								

wall time: 1 s, 10 s, 100 s, 1,000 s